NASATurbofanLevelB2 (A level upgrade): Initial model results

- codes: https://github.com/drvojtex/ADAML_NASATurbofanLevelB2

Goals

Filling in missing data using multivariate approaches. One of the sensors has stopped emitting. Choose one sensor. How well can you estimate its values, using multivariate regression approaches? What is the appropriate number of latent variables needed, and what is the minimum number of sensors that are used to predict the faulty sensor?

Initial experimental evaluation procedure

The aim of the initial experimental evaluation was primarily to estimate the approximate number of latent variables required for the PCR and PLS models and to assess whether it is possible to identify the trend of the missing sensor's measurements based on a limited subset of sensors providing input data to the model.

Model calibration, validation, and testing were conducted using a combination of fixed data splits and three-fold cross-validation. The dataset was divided into five parts, with parts 1–3 used for training and calibration, part 4 for validation, and part 5 for final testing. The input data were zscore standardized and to the input windows was added information about context (see Part 2). During the cross validation, the models parameters were set based on the training data and the best-performing fold was selected based on the calibration data. The optimal hyperparameters (k – number of latent variables, w – window size, D – set of sensors) were selected based on the validation Q^2, and the final model performance was evaluated on the test set using Q^2, and RMSE.

The initial experimental evaluation was performed by setting the hyperparameters window size (w) to 5 and 10, and the number of latent variables (k) to 3 and 5. The evaluation was carried out using all non-constant sensors (excluding the target sensor) as well as a subset of two sensors that, according to the biplot analysis (see Part 1), were expected to carry the most information in common with the target sensor. Tables A1, A2, A3, and A4 present the results of this initial experimental evaluation for operational settings (OpS) 1 and 3, where in operational setting 1 the missing target sensor was S17, and in operational setting 3 the target (missing) sensor was S21. For both operational settings, PCR and PLS models (mathematical description see Part 3) were experimentally evaluated. The best performance was achieved for a window size of 5 and five latent variables when all sensors were included. This finding can thus be used to guide further analysis of the appropriate number of latent variables, indicating that the optimal number of latent variables and window size is likely below 10.

The Q^2 on testing data were as follows. OpS 1: PCR 0.59, PLS 0.59; OpS 3: PCR 0.83, PLS 0.83.

It is also important to note that although regression based on only two input sensors (operational setting 1 – sensors 3 and 11; operational setting 3 – sensors 7 and 20) produced slightly worse results (in terms of RMSE and Q^2 on validation data), the performance degradation was not substantial, suggesting that even two sensors are sufficient to identify the trend of the missing sensor.

Analysis of latent number of variables

Tables 1 show Q^2 values on validation data depending on the number of latent variables. These results were obtained for both OpS 1 and OpS 3 with a window size of 5 on the best-performing folds identified during the initial experimental evaluations. The results show that the PLS model required fewer latent variables to achieve better performance compared to PCR, especially in OpS 1. Table 2 shows that PLS exhibits lower cumulative explained variance in the training data than PCR, as it prioritizes components that best predict the target sensor values rather than those explaining the largest variance in the input sensor values. Both PCR and PLS models exhibit a gradual increase in explained variance in training data as the number of latent variables increases. For OpS 1, the variance grows only slightly beyond three components, matching the plateau observed in Q^2 values. For OpS 3, the variance continues to increase more noticeably, in line with the improvement of Q^2. Overall, both models show stable behaviour after several latent variables, confirming that the main structure of the data is captured within the first few components. These results suggest that both models effectively represent the underlying data patterns with a limited number of latent variables, with PLS showing slightly higher efficiency. The consistent trends between explained variance and Q^2 further indicate good model regularization and support the selection of a small number of latent variables for optimal complexity and generalization.

Table 1 – Q2 values on validation data (dependency on number of latent variables)									
Latent variables	1	2	3	4	5	6	7	8	9
OpS1, PCR, target 17, w 5, fold 3, all sensors	0.5769	0.6041	0.6041	0.6042	**0.6043**	0.6043	0.6042	0.6041	0.6040
OpS1, PLS, target 17, w 5, fold 3, all sensors	0.5879	**0.6043**	0.6043	0.6043	0.6021	0.6014	0.6008	0.6007	0.6004
OpS3, PCR, target 21, w 5, fold 1, all sensors	0.1685	0.7706	0.8103	0.8246	0.8328	0.8330	0.8332	**0.8333**	0.8333
OpS3, PLS, target 21, w 5, fold 1, all sensors	0.7374	0.7996	0.8297	0.8316	0.8322	0.8328	0.8343	**0.8351**	0.8350

Table 2 – cumulative explained variance in training data (dependency on number of latent variables) [%]									
Latent variables	1	2	3	4	5	6	7	8	9
OpS1, PCR, target 17, w 5, fold 3, all sensors	63.45	76.71	77.74	78.66	79.43	80.16	80.84	81.43	82.01
OpS1, PLS, target 17, w 5, fold 3, all sensors	63.39	76.71	77.48	78.18	78.58	79.03	79.45	79.99	80.62
OpS3, PCR, target 21, w 5, fold 1, all sensors	49.96	73.19	78.64	83.41	85.04	85.87	86.65	87.36	88.00
OpS3, PLS, target 21, w 5, fold 1, all sensors	35.10	73.08	78.25	80.36	84.60	85.52	85.98	87.07	87.07

Results visualisation and residuals analysis

As a next step, there was performed an analysis on residuals for the best-performing configurations (given by the previous section) of models. Figures 1, 2, 3 and 4 show example comparisons of predictions and true signal data. It can be seen that the trend of the missing sensors is well captured, however the high-frequency component is not captured. The question is: is it a high-frequency noise or a periodical component?

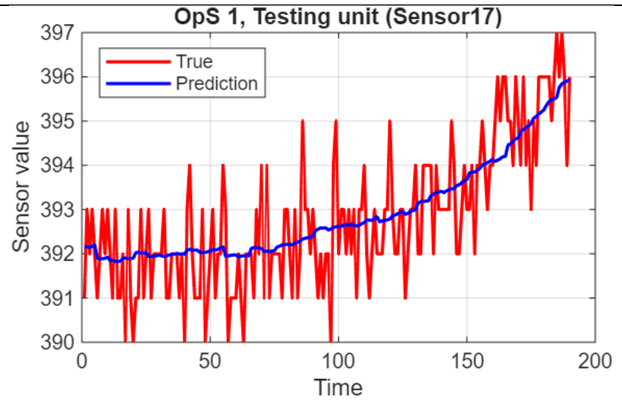

Figure 1 – PCR, OpS 1

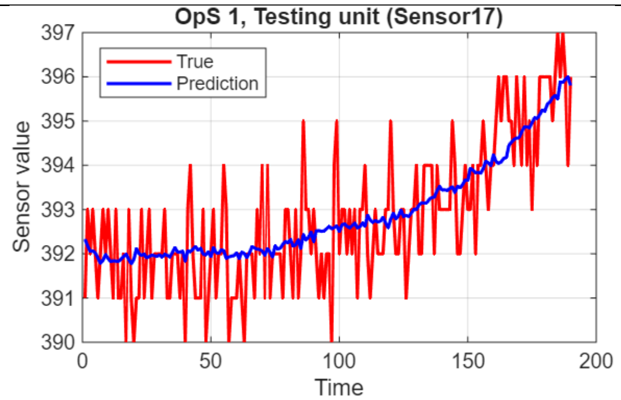

Figure 2 – PLS, OpS 1

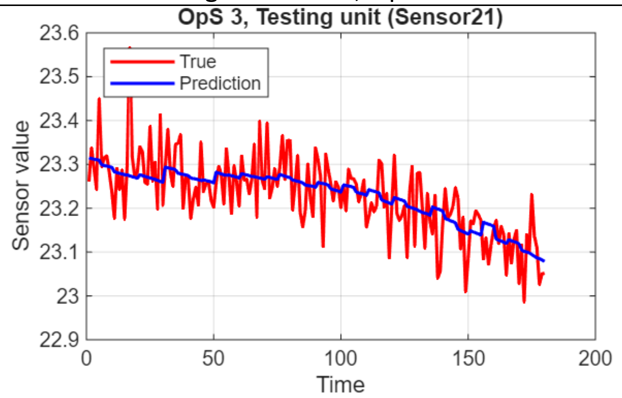

Figure 3 – PCR, OpS 3

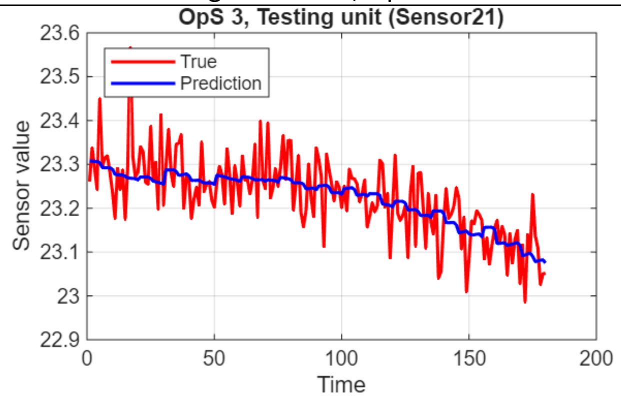

Figure 4 – PLS, OpS 3

To answer such a question about the origin of residuals, an autocorrelation analysis was performed. Figures 5, 6, 7 and 8 show autocorrelation plots for corresponding samples predictions and true sensor data from previous Figures. It is visible that for lags greater than zero, the autocorrelation function is bounded between -0.2 and 0.2.

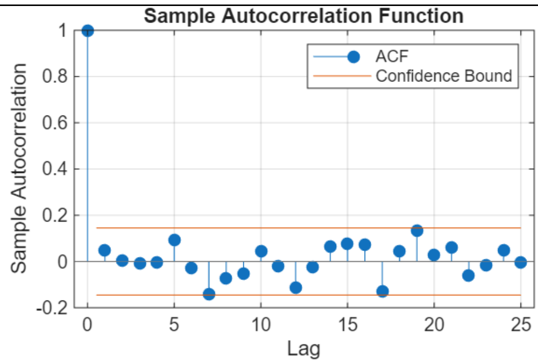

Figure 5 – PCR, OpS 1

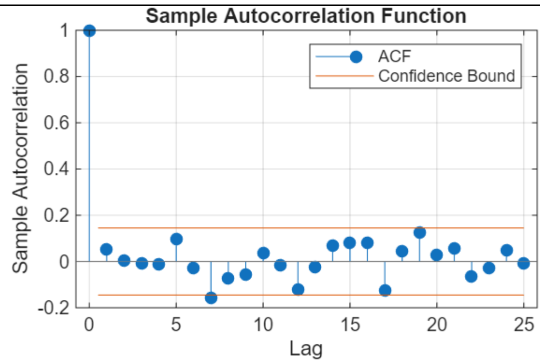

Figure 6 – PLS, OpS1

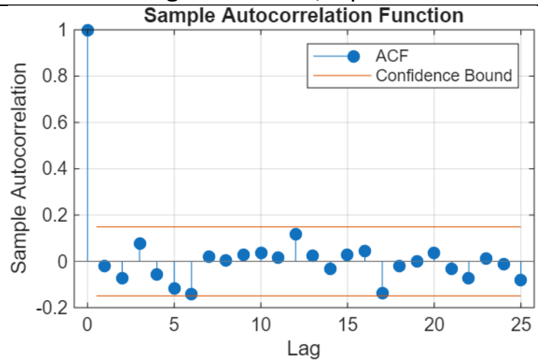

Figure 7 – PCR, OpS 3

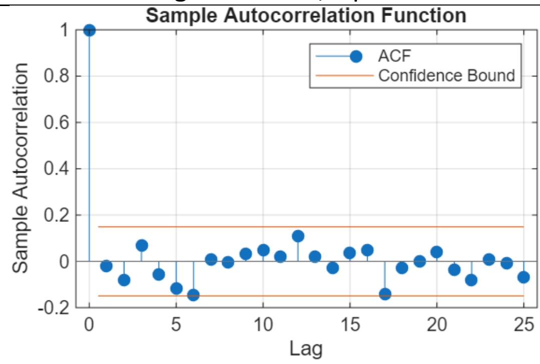

Figure 8 – PLS, OpS 3

Figures 9, 10, 11 and 12 show the minimum and maximum values of each lag of autocorrelation functions computed over all testing Units. The values of lags (greater or equal one) are bounded between -0.2 and 0.2, and their average values are approximately equal zero. From this can be concluded that the residuals behave like white noise and the models are capturing the information of the target (missing) sensor.

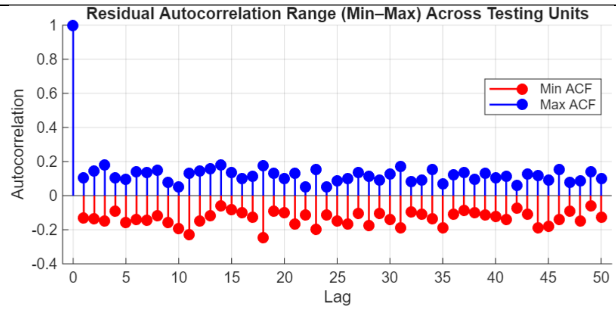

Figure 9 – PCR, OpS 1

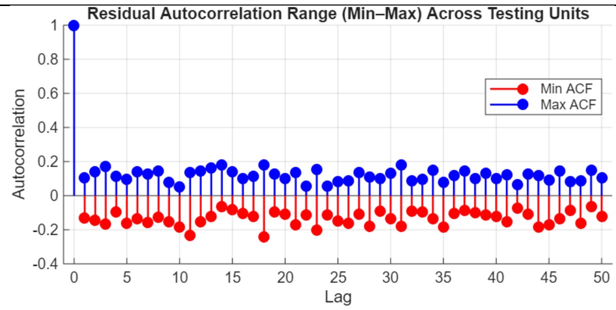

Figure 10 – PLS, OpS 1

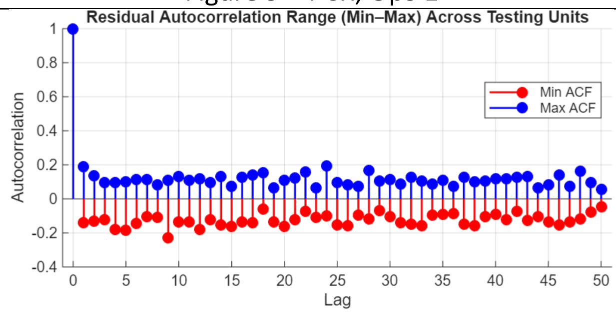

Figure 11 – PCR, OpS 3

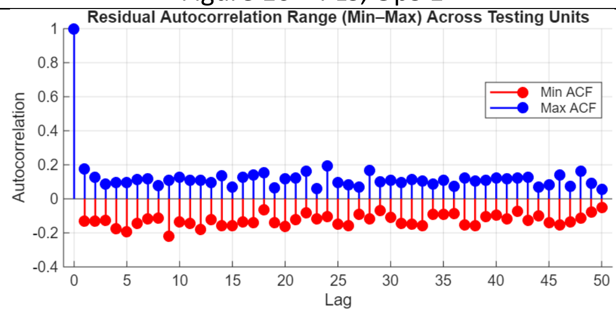

Figure 12 – PLS, OpS 3

Table 3 summarizes the results of statistical tests on residuals for the PCR and PLS models across OpS 1 and 3. The Ljung–Box tests [1] assessed autocorrelation in the residuals at several lag values and showed that the majority of tested units did not reject the null hypothesis, indicating that the residuals are largely uncorrelated. The number of failures was low, typically 1–2 out of 20 units, with minimal differences between models and operating states. The ARCH tests [2], which check for constant variance in the residuals, showed zero or very few failures, indicating that the residual variance is stable and heteroskedasticity is not a significant concern. Overall, the residuals of both models behave approximately as white noise, supporting the correct specification of the models. The results are consistent across PCR and PLS models and across different OpS, suggesting robustness. Individual minor failures can be considered random deviations and do not represent a systematic issue.

Table 3 – number of testing Units rejecting H_0 of statistical tests on residuals (alpha = 0.05)					
Model, OpS	Ljung-Box (5 Lags)	Ljung-Box (10 Lags)	Ljung-Box (20 Lags)	Ljung-Box (20 Lags)	ARCH test
PCR, 1	1/20	1/20	2/20	2/20	0/20
PLS, 1	1/20	1/20	2/20	2/20	0/20
PCR, 3	2/20	2/20	1/20	0/20	1/20
PLS, 3	2/20	2/20	1/20	0/20	1/20

Figures 13, 14, 15 and 16 show scatter plots of the dependency of predictions on true values. It can be visible that for the OpS 1 the target sensor S17 does not produce continuous values, however it changes its measurements in steps. This could be the reason for the worse performance than the prediction of S21 in OpS 3.

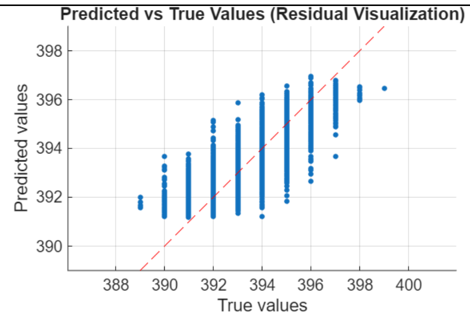

Figure 13 – PCR, OpS 1

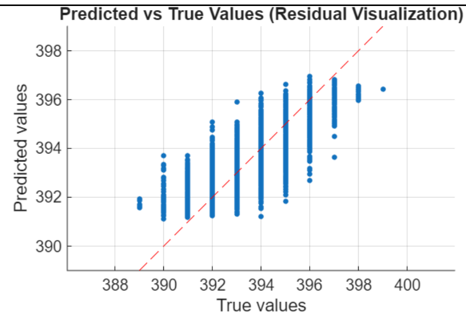

Figure 14 – PLS, OpS 1

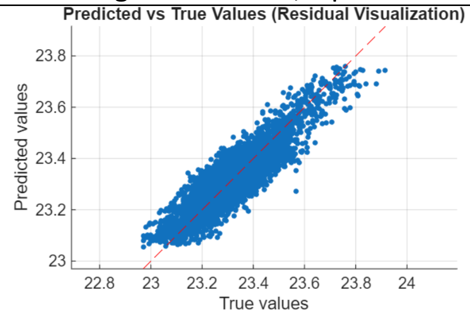

Figure 15 – PCR, OpS 3

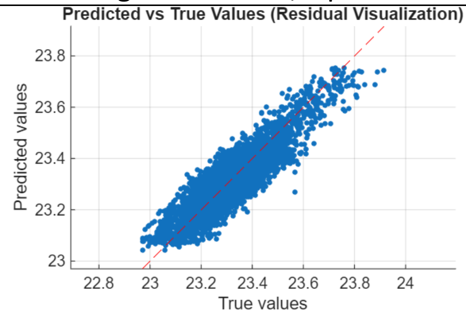

Figure 16 – PLS, OpS 3

Conclusion

Both PLS and PCR were able to predict the trend of the missing sensor using data from the other sensors. The analysis of residuals showed that the part of the signal not captured by the model behaves like white noise, which means the models fit the data well. The trend of the target sensor can be estimated accurately even when using data from only two sensors that share the most information with it. The models also do not need many latent variables — in particular, the PLS model in OpS 1 can estimate the trend well with just two latent components. Overall, the PLS and PCR models offer a simple and effective way to estimate the trends of missing sensors.

Author's contributions

Jarkko Komulainen created functions utils/R2eval.m, utils/autocorrEval.m and utils/resAnalysis.m. The rest, including reporting, was done by Vojtěch Drahý.

References

[1] 10.1093/biomet/65.2.297

[2] 10.2307/1912773

The Grammarly tool was used while writing the report. [https://app.grammarly.com/]

Table A1 – Operational setting 1 (target Sensor 17), PCR					
Hyperparameters setup	Fold	Training (RMSE, R2)	Calibration (RMSE, Q2)	Validation (RMSE, Q2)	Testing (RMSE, Q2)
window size = 5 k = 3 all sensors	1	0.9478, 0.6099	0.9470, 0.6105		
	2	0.9459, 0.6160	0.9508, 0.5996		
	3	0.9486, 0.6037	0.9454, **0.6232**	0.9716, 0.6041	
window size = 5 k = 5 all sensors	1	0.9472, 0.6105	0.9472, 0.6104		
	2	0.9455, 0.6163	0.9505, 0.5999		
	3	0.9483, 0.6040	0.9449, **0.6236**	0.9713, **0.6043**	0.9643, 0.5899
window size = 10 k = 3 all sensors	1	0.9463, 0.5973	0.9461, 0.5957		
	2	0.9451, 0.6018	0.9484, 0.5881		
	3	0.9467, 0.5905	0.9456, **0.6092**	0.9715, 0.5865	
window size = 10 k = 5 all sensors	1	0.9462, 0.5974	0.9462, 0.5956		
	2	0.9450, 0.6019	0.9483, 0.5881		
	3	0.9465, 0.5907	0.9457, **0.6091**	0.9715, 0.5866	
window size = 5 k = 3 S 3, 11	1	0.9692, 0.5922	0.9670, 0.5940		
	2	0.9666, 0.5990	0.9719, 0.5817		
	3	0.9689, 0.5866	0.9674, **0.6055**	1.001, 0.5800	
window size = 5 k = 5 S 3, 11	1	0.9690, 0.5924	0.9672, 0.5938		
	2	0.9666, 0.5990	0.9720, 0.5816		
	3	0.9688, 0.5867	0.9674, 0.**6056**	1.001, 0.5799	
window size = 10 k = 3 S 3, 11	1	0.9608, 0.5849	0.9568, 0.5865		
	2	0.9584, 0.5906	0.9616, 0.5766		
	3	0.9586, 0.5802	0.9614, **0.5962**	0.9901, 0.5709	
window size = 10 k = 5 S 3, 11	1	0.9608, 0.5850	0.9568, 0.5866		
	2	0.9583, 0.5907	0.9619, 0.5763		
	3	0.9586, 0.5803	0.9615, **0.5961**	0.9902, 0.5707	

Table A2 – Operational setting 1 (target Sensor 17), PLS					
Hyperparameters setup	Fold	Training (RMSE, R2)	Calibration (RMSE, Q2)	Validation (RMSE, Q2)	Testing (RMSE, Q2)
window size = 5 k = 3 all sensors	1	0.9470, 0.6106	0.0471, 0.6105		
	2	0.9482, 0.6141	0.9538, 0.5971		
	3	0.9478, 0.6044	0.9451, **0.6234**	0.9714, **0.6043**	0.9641, 0.5901
window size = 5 k = 5 all sensors	1	0.9453, 0.6120	0.9480, 0.6097		
	2	0.9462, 0.6157	0.9553, 0.5958		
	3	0.9460, 0.6059	0.9464, **0.6224**	0.9740, 0.6021	
window size = 10 k = 3 all sensors	1	0.9462, 0.5974	0.9466, 0.5953		
	2	0.9503, 0.5973	0.9548, 0.5824		
	3	0.9462, 0.5910	0.9465, **0.6085**	0.9725, 0.5857	
window size = 10 k = 5 all sensors	1	0.9444, 0.5989	0.9482, 0.5939		
	2	0.9484, 0.5989	0.9567, 0.5807		
	3	0.9445, 0.5924	0.9477, **0.6075**	0.9744, 0.5841	
window size = 5 k = 3 S 3, 11	1	0.9673, 0.5938	0.9634, 0.5970		
	2	0.9668, 0.5988	0.9720, 0.5816		
	3	0.9656, 0.5894	0.9670, **0.6059**	0.9988, 0.5817	
window size = 5 k = 5 S 3, 11	1	0.9670, 0.5941	0.9636, 0.5968		
	2	0.9666, 0.5990	0.9724, 0.5813		
	3	0.9654, 0.5896	0.9669, **0.6059**	0.9991, 0.5815	
window size = 10 k = 3 S 3, 11	1	0.9608, 0.5850	0.9576, 0.5859		
	2	0.9624, 0.5872	0.9652, 0.5734		
	3	0.9580, 0.5808	0.9614, **0.5962**	0.9896, 0.5713	
window size = 10 k = 5 S 3, 11	1	0.9603, 0.9582	0.9582, 0.5853		
	2	0.9619, 0.5876	0.9655, 0.5731		
	3	0.9578, 0.5810	0.9614, **0.5963**	0.9900, 0.5709	

Table A3 – Operational setting 3 (target Sensor 21), PCR					
Hyperparameters setup	Fold	Training (RMSE, R2)	Calibration (RMSE, Q2)	Validation (RMSE, Q2)	Testing (RMSE, Q2)
window size = 5 k = 3 all sensors	1	0.0632, 0.8065	0.0631, **0.8266**	0.0659, 0.8103	
	2	0.0631, 0.8186	0.0612, 0.8139		
	3	0.0622, 9.8197	0.0645, 0.8040		
window size = 5 k = 5 all sensors	1	0.0612, 0.8181	0.0612, **0.8328**	0.0619, **0.8328**	0.0608, 0.8332
	2	0.0615, 0.8279	0.0606, 0.8176		
	3	0.0608, 0.8274	0.0619, 0.8199		
window size = 10 k = 3 all sensors	1	0.0625, 0.8020	0.0622, **0.8238**	0.0641, 0.8143	
	2	0.0626, 0.8141	0.0608, 0.8079		
	3	0.0612, 0.8175	0.0634, 0.8033		
window size = 10 k = 5 all sensors	1	0.0610, 0.8114	0.0609, **0.8313**	0.0616, 0.8283	
	2	0.0612, 0.8219	0.0604, 0.8103		
	3	0.0606, 0.8208	0.0617, 0.8277		
window size = 5 k = 3 S 7, 20	1	0.0654, 0.7919	0.0657, **0.8122**	0.0662, 0.8084	
	2	0.0659, 0.8025	0.0648, 0.7914		
	3	0.0652, 0.8021	0.0662, 0.7935		
window size = 5 k = 5 S 7, 20	1	0.0652, 0.7936	0.0657, **0.8130**	0.0660, 0.8098	
	2	0.0657, 0.8034	0.0645, 0.7935		
	3	0.0650, 0.8037	0.0661, 0.7944		
window size = 10 k = 3 S 7, 20	1	0.0634, 0.7964	0.0634, **0.8174**	0.0640, 0.8146	
	2	0.0637, 0.8071	0.0626, 0.7961		
	3	0.0630, 0.8068	0.0642, 0.7982		
window size = 10 k = 5 S 7, 20	1	0.0634, 0.7970	0.0634, **0.8176**	0.0640, 0.8148	
	2	0.0637, 0.8074	0.0626, 0.7996		
	3	0.0630, 0.8073	0.0642, 0.7984		

Table A4 – Operational setting 3 (target Sensor 21), PLS					
Hyperparameters setup	Fold	Training (RMSE, R2)	Calibration (RMSE, Q2)	Validation (RMSE, Q2)	Testing (RMSE, Q2)
window size = 5 k = 3 all sensors	1	0.0618, 0.8134	0.0617, **0.8343**	0.0624, 0.8297	
	2	0.0619, 0.8254	0.0607, 0.8170		
	3	0.0613, 0.8246	0.0629, 0.8139		
window size = 5 k = 5 all sensors	1	0.0712, 0.8177	0.0613, **0.8363**	0.0620, **0.8322**	0.0609, 0.8328
	2	0.0614, 0.8286	0.0605, 0.8183		
	3	0.0610, 0.8268	0.0620, 0.8192		
window size = 10 k = 3 all sensors	1	0.0617, 0.8071	0.0615, **0.8279**	0.0623, 0.8246	
	2	0.0617, 0.8191	0.0605, 0.8096		
	3	0.0611, 0.8180	0.0627, 0.8077		
window size = 10 k = 5 all sensors	1	0.0612, 0.8100	0.0611, **0.8301**	0.0619, 0.8269	
	2	0.0612, 0.8221	0.0604, 0.8104		
	3	0.0608, 0.8196	0.0619, 0.8127		
window size = 5 k = 3 S 7, 20	1	0.0656, 0.7908	0.0658, **0.8118**	0.0666, 0.8077	
	2	0.0659, 0.8025	0.0648, 0.7915		
	3	0.0656, 0.7993	0.0666, 0.7914		
window size = 5 k = 5 S 7, 20	1	0.0653, 0.7930	0.0655, **0.8134**	0.0660, 0.8096	
	2	0.0655, 0.8044	0.0644, 0.7940		
	3	0.0652, 0.8016	0.0663, 0.7932		
window size = 10 k = 3 S 7, 20	1	0.0636, 0.7948	0.0635, **0.8165**	0.0642, 0.8136	
	2	0.0638, 0.8069	0.0627, 0.7958		
	3	0.0634, 0.8039	0.0646, 0.7959		
window size = 10 k = 5 S 7, 20	1	0.0635, 0.7953	0.0635, **0.8166**	0.06415, 0.8138	
	2	0.0636, 0.8077	0.0626, 0.7966		
	3	0.0633, 0.8046	0.0645, 0.7963		